Project 3: Bayesian Inference and MCMC

Math 540: Uncertainty Quantification

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Problem 1: Heat Model Bayesian Inference (Copper Rod)

Problem Statement

Use the MATLAB code to repeat the analysis in Example 12.17 for the copper rod data. Use $k = 4.01$ for the thermal conductivity and $u_{amb} = 22.28$. Compare the chains and marginal distributions computed using the Metropolis Algorithm 12.9 with those generated by DRAM. Compare your results with the frequentist estimates from Project 2.

Solution

From the previous project, we obtained optimal parameter values using the Frequentist approach. To apply the Metropolis Algorithm, we use the covariance matrix:

$$V = \begin{bmatrix} 0.0090 & -1.2117 \times 10^{-6} \\ -1.2117 \times 10^{-6} & 1.7726 \times 10^{-10} \end{bmatrix}$$

As described in Metropolis Algorithm 12.9, we use $R = \text{chol}(V)$ to define the proposal distribution. For faster convergence, we initialize with the OLS optimal parameters: $\Phi = -9.9265$ and $h = 0.0014$.

MCMC Results

The acceptance probability for the Metropolis algorithm is:

$$P(\text{accept } \theta^*) = 0.2701$$

DRAM Results

The chain covariance matrix using DRAM algorithm is:

$$V = \begin{bmatrix} 0.01127 & -1.5201 \times 10^{-6} \\ -1.5201 \times 10^{-6} & 2.2261 \times 10^{-10} \end{bmatrix}$$

This is very close to the initial (Frequentist) covariance matrix.

Comparison of Methods

Method	h	Φ	σ	σ_h	σ_Φ
OLS	0.0014	-9.9265	0.2301	0.0947	1.3314×10^{-5}
MCMC	0.0014	-9.9281	0.2542	0.0105	1.4716×10^{-5}
DRAM	0.0014	-9.9285	0.2556	0.0106	1.4920×10^{-5}

Method	h	Φ	σ	σ_h	σ_Φ

Table 1: Parameter estimations and standard deviations using OLS, MCMC and DRAM algorithms.

The parameter estimates are close across all methods. The standard deviations from Bayesian approaches are slightly larger than Frequentist estimates.

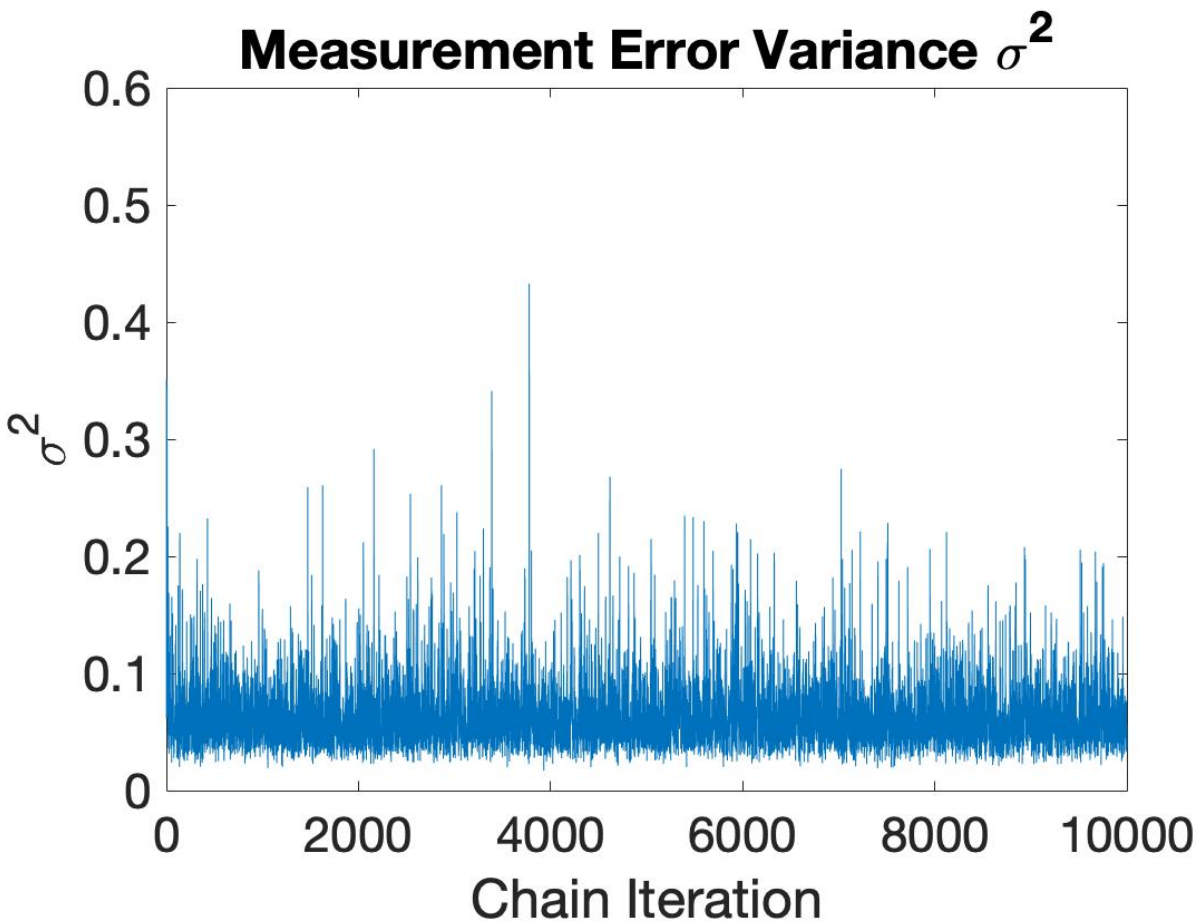

Figure 1: Chain Plots

Figure 1: Chain plots and joint samples using DRAM and Metropolis Algorithm.

Figure 2: Marginal posterior distributions comparing DRAM and MCMC.

Problem 2: SIR Model Bayesian Inference

Problem Statement

Consider the SIR model:

$$\frac{dS}{dt} = \delta N - \delta S - \gamma IS, \quad S(0) = 900$$

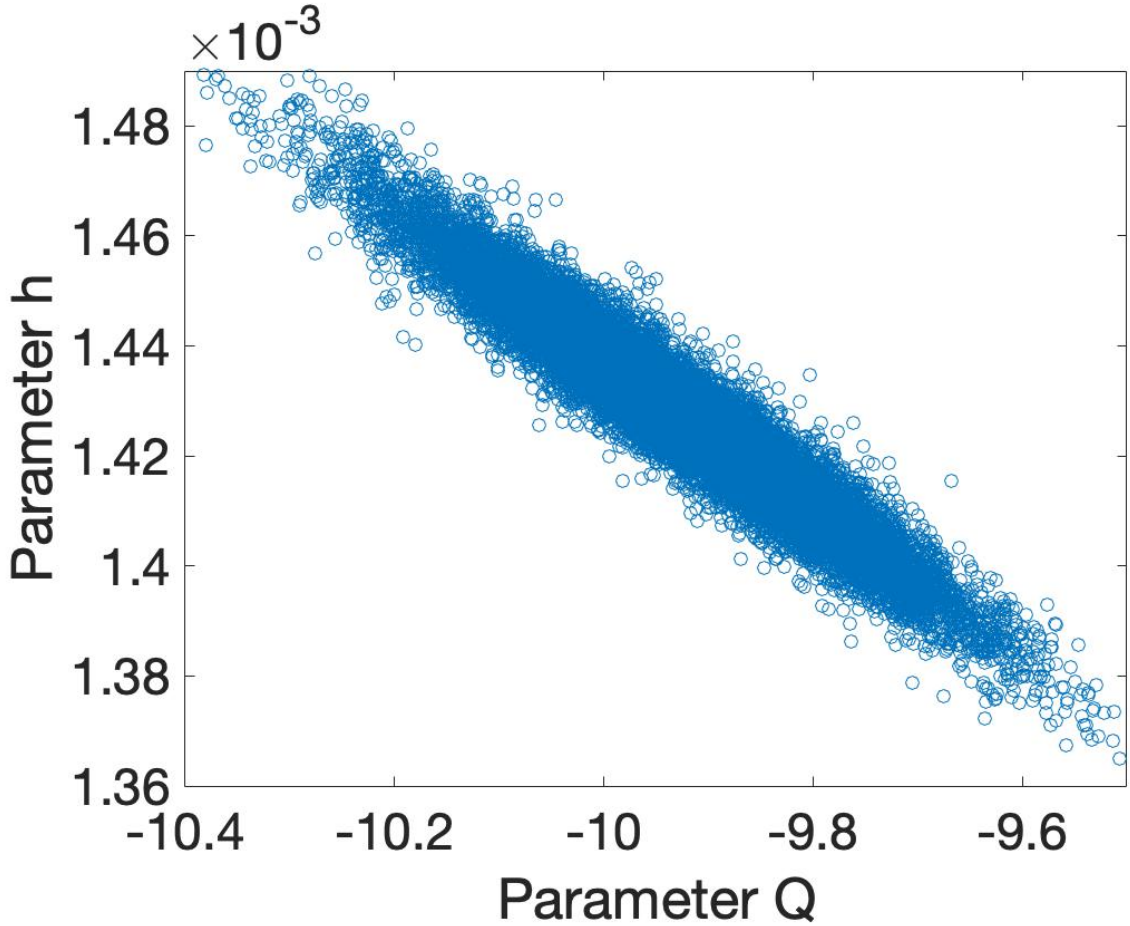

Figure 2: Joint Samples MCMC

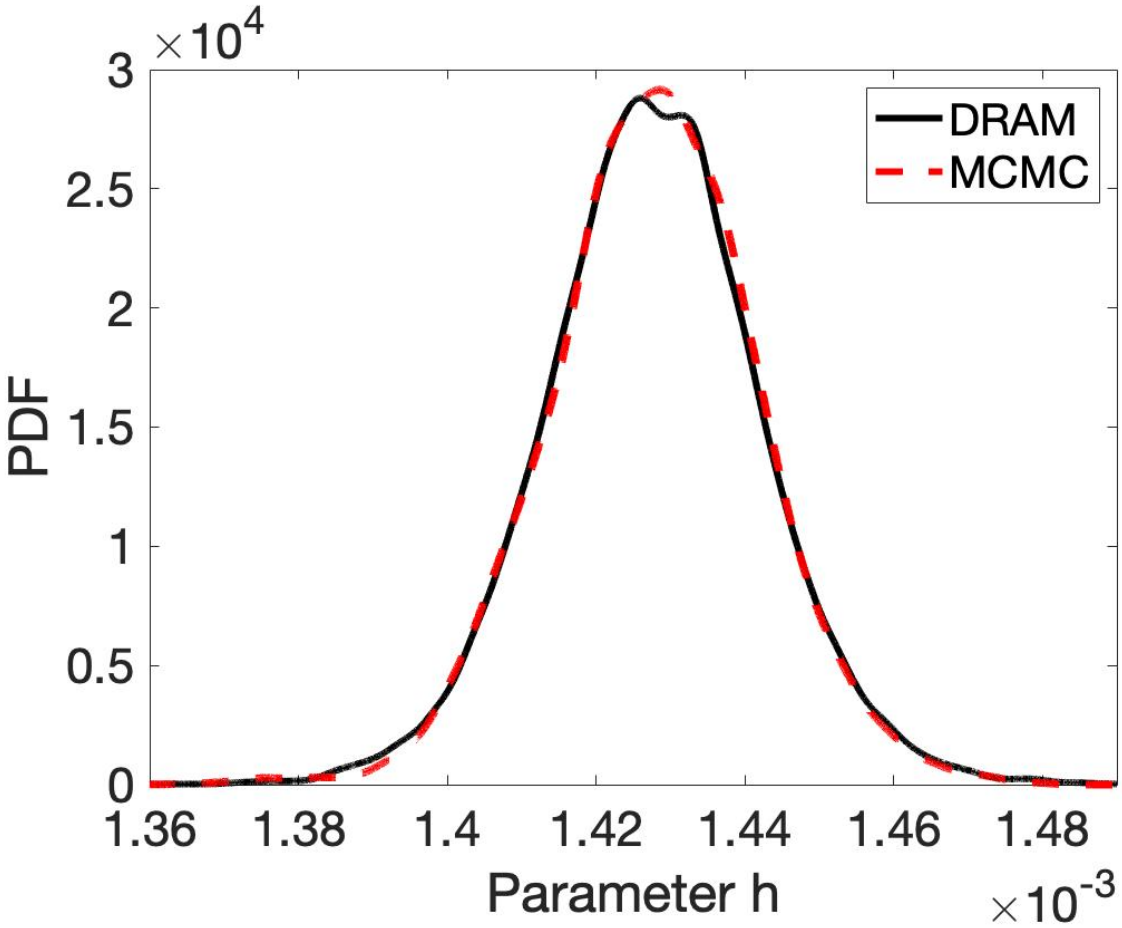

Figure 3: Joint Samples DRAM

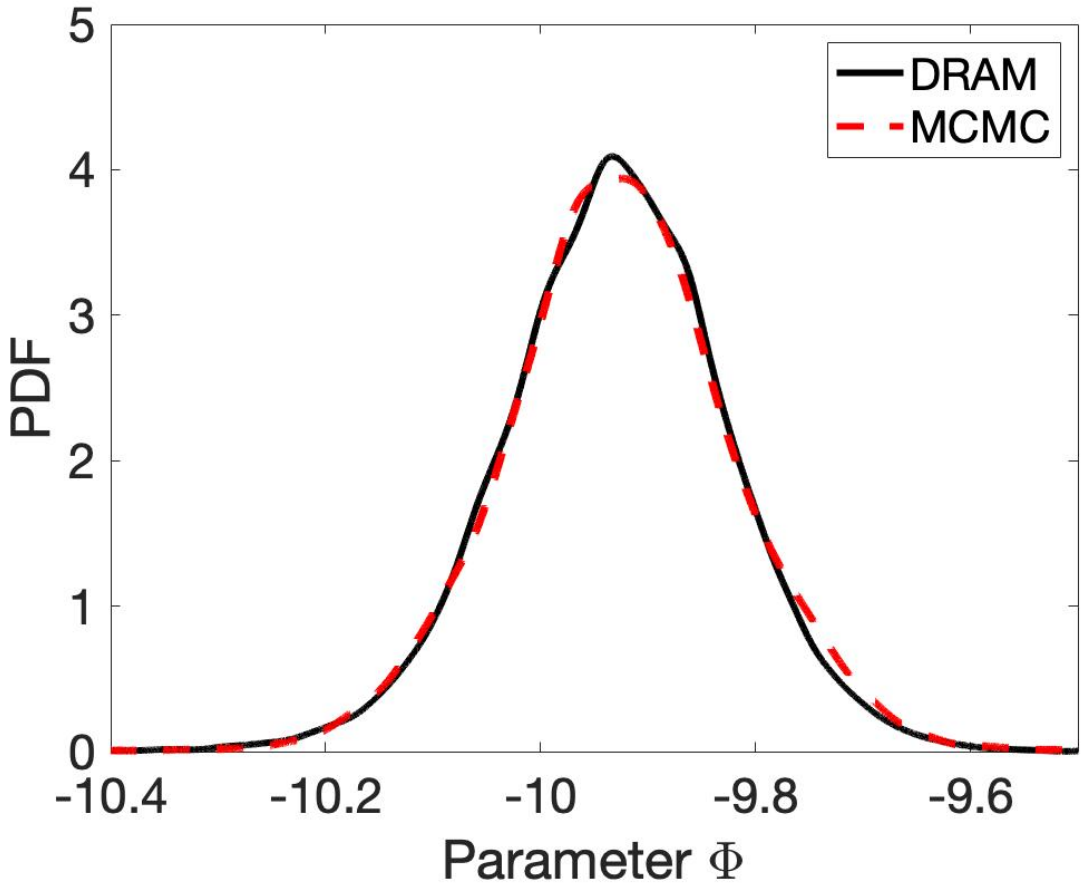

Figure 4: Posterior Distributions

$$\frac{dI}{dt} = \gamma IS - (r+\delta)I, \quad I(0) = 100$$

$$\frac{dR}{dt} = rI - \delta R, \quad R(0) = 0$$

where $\gamma, r, \delta \in [0, 1]$.

Part (a): Three-Parameter Model

The OLS estimated parameters are used as initial values for DRAM:

$$\gamma = 0.0100, \quad \delta = 0.1953, \quad r = 0.7970$$

Initial covariance matrix:

$$V = \begin{bmatrix} 1.7357 \times 10^{-8} & 1.8575 \times 10^{-7} & 1.4540 \times 10^{-7} \\ 1.8575 \times 10^{-7} & 2.6153 \times 10^{-5} & 3.1790 \times 10^{-5} \\ 1.4540 \times 10^{-7} & 3.1790 \times 10^{-5} & 8.4797 \times 10^{-5} \end{bmatrix}$$

Estimated observational standard deviation: $\sigma = 21.0528$ (Bayesian) vs $\sigma = 20.6586$ (OLS)

Chain covariance matrix:

$$V = \begin{bmatrix} 4.9214 \times 10^{-8} & 8.2561 \times 10^{-8} & 1.3512 \times 10^{-7} \\ 8.2561 \times 10^{-8} & 1.0218 \times 10^{-4} & 1.1635 \times 10^{-4} \\ 1.3512 \times 10^{-7} & 1.1635 \times 10^{-4} & 2.1544 \times 10^{-4} \end{bmatrix}$$

Bayesian parameter estimates:

$$\gamma = 0.0100, \quad \delta = 0.1958, \quad r = 0.7979$$

These are close to the Frequentist estimates.

Figure 3: Chain plots, pairwise samples, and variance chain for SIR model.

Figure 4: Marginal posterior distributions comparing DRAM and OLS.

Part (b): Non-Identifiable Four-Parameter Model

Consider the model with parameters $\theta = [\gamma, r, \delta, k]$ where the model includes the product γk:

$$\frac{dS}{dt} = \delta N - \delta S - \gamma k IS$$

Without specifying an initial covariance matrix, we obtain non-converged chains. The joint sample distribution shows that $\gamma \times k$ equals a constant, revealing the non-identifiability.

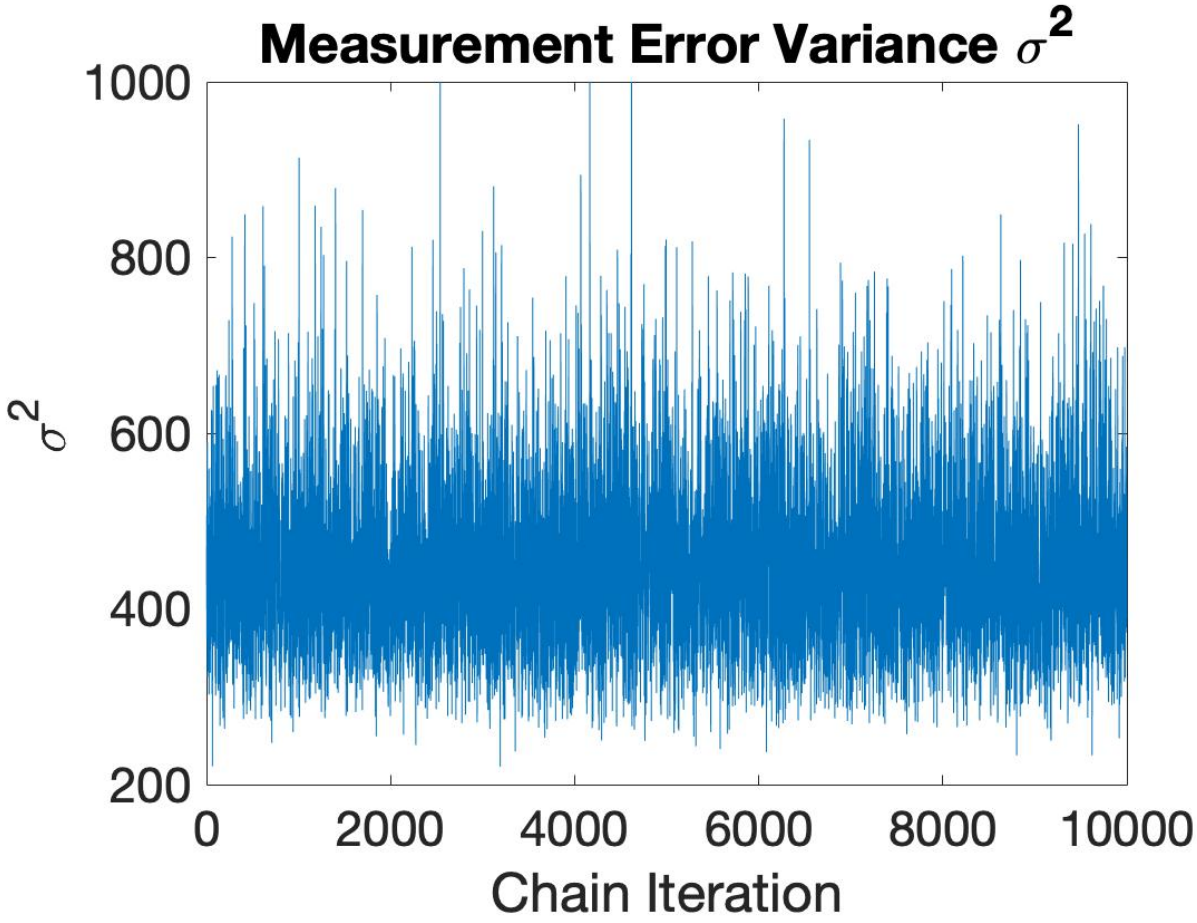

Figure 5: SIR Chain Plots

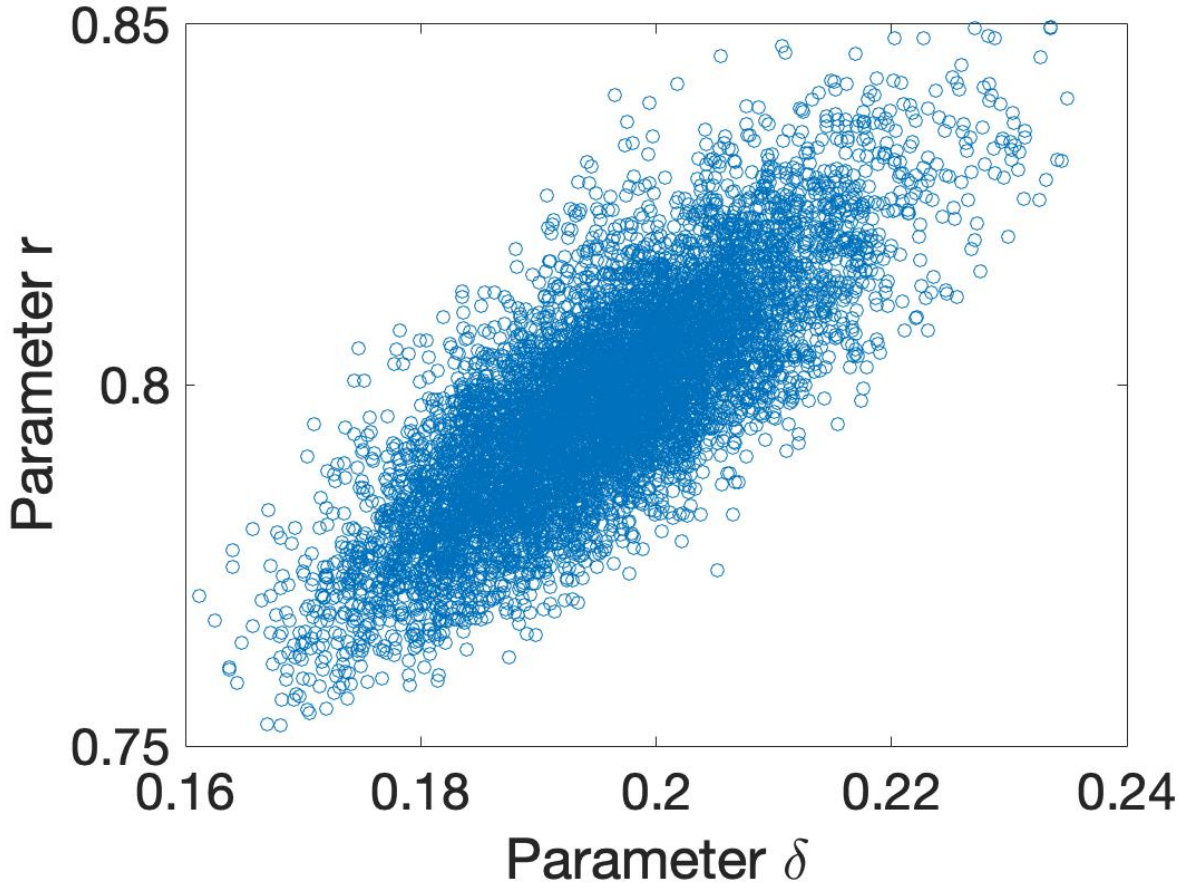

Figure 6: SIR Pairwise

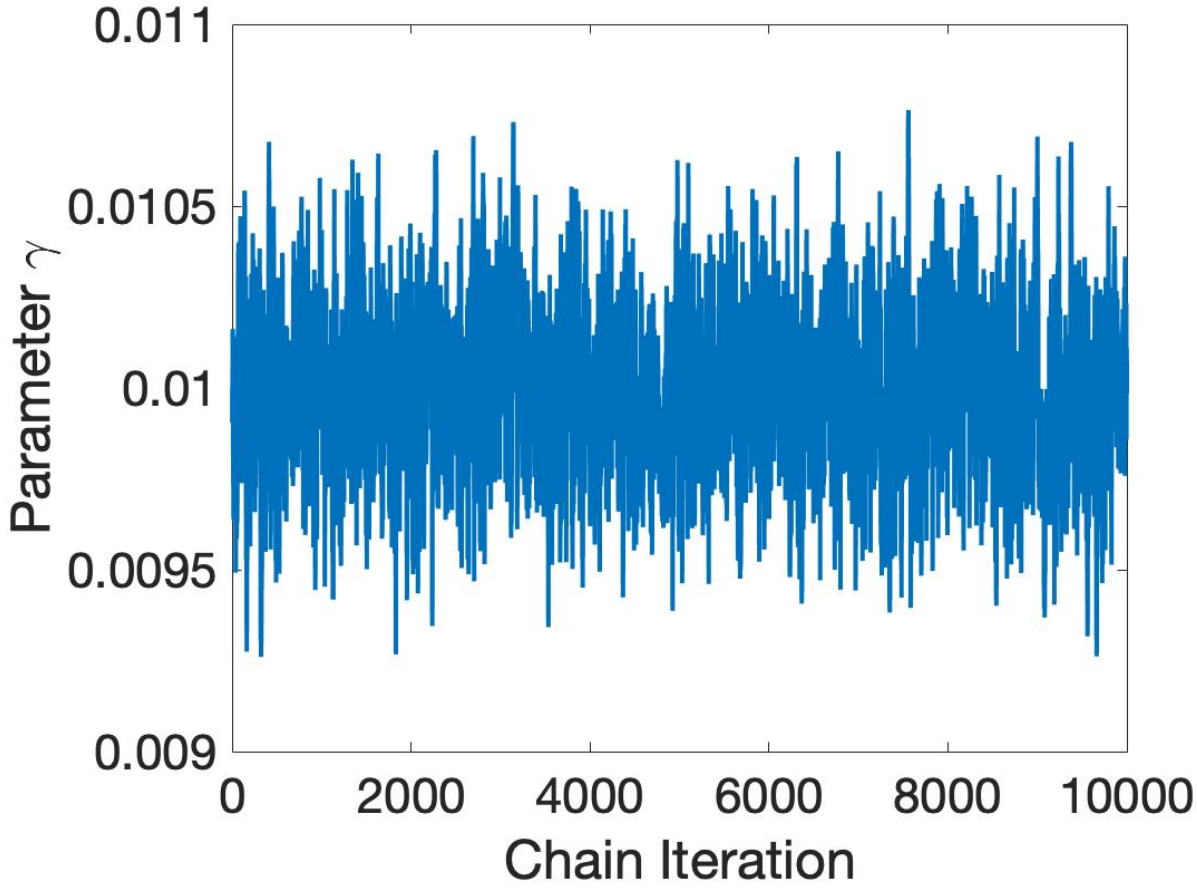

Figure 7: SIR Variance Chain

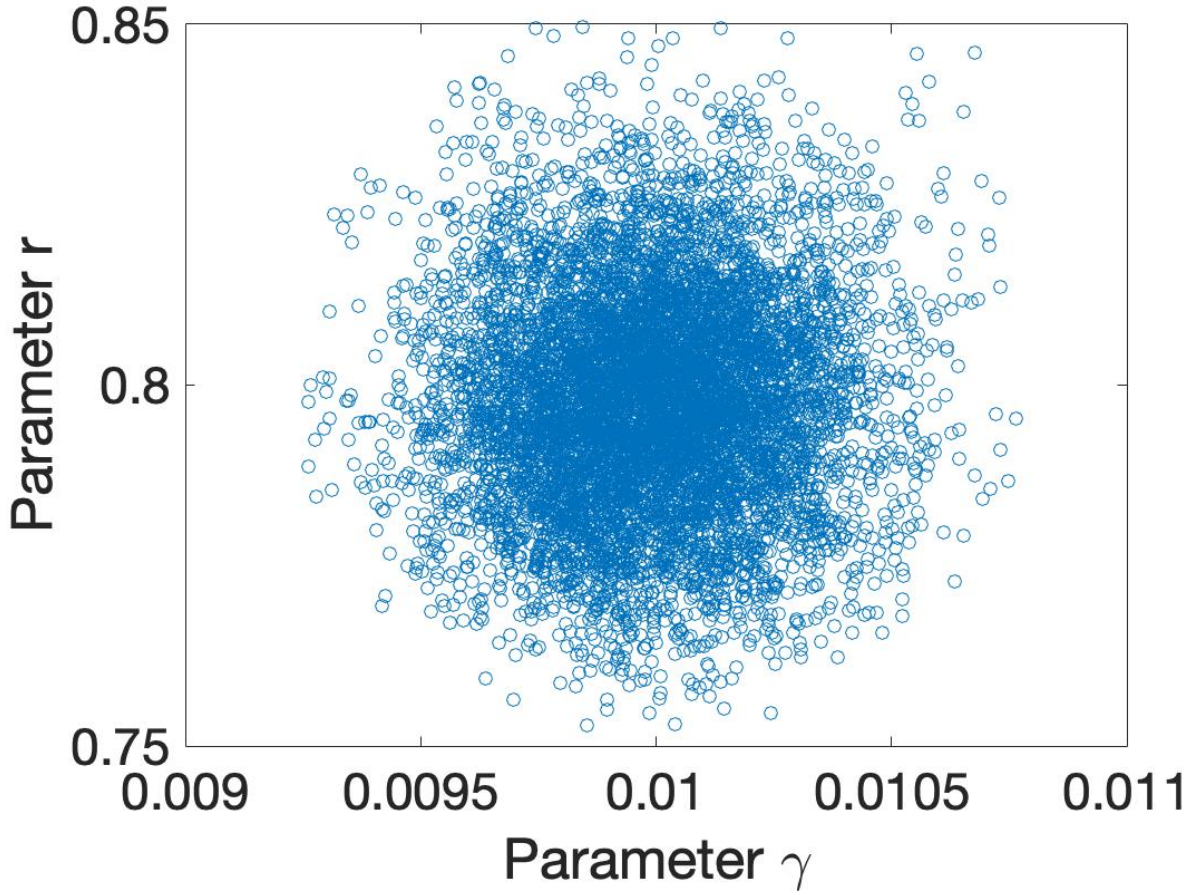

Figure 8: SIR Marginal Distributions

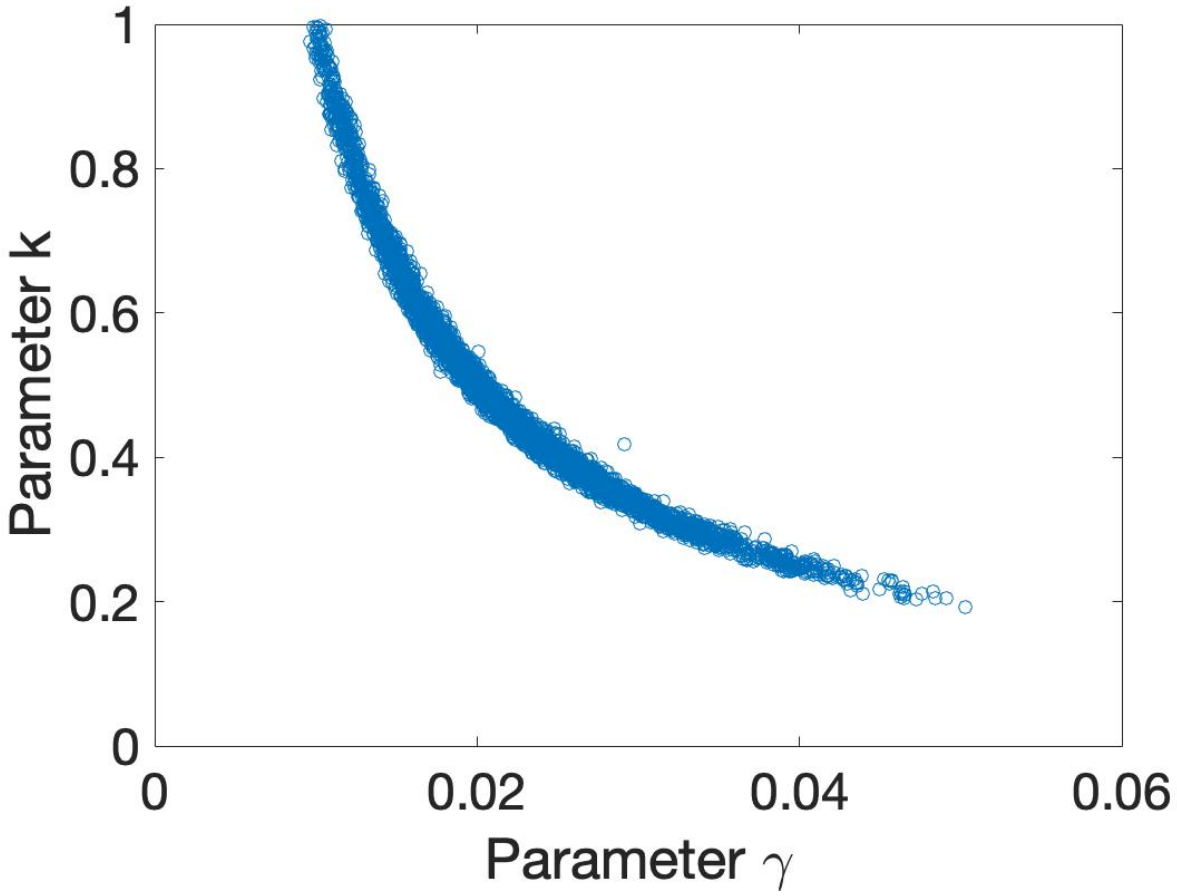

Figure 9: Non-identifiable Chains

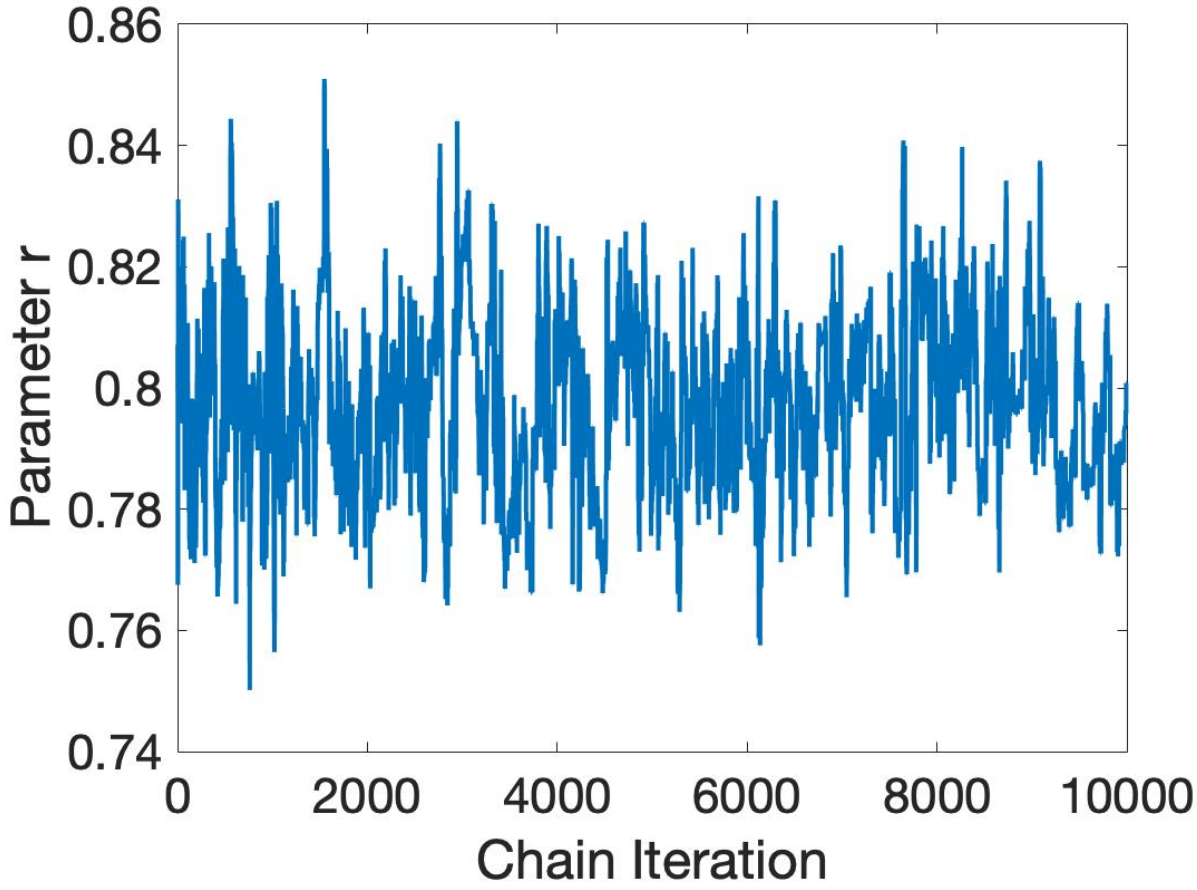

Figure 10: Non-identifiable Joint

Key Finding: Bayesian analysis can identify mutually non-identifiable parameters through the pairwise posterior distributions.

Figure 5: Non-converged chains and joint samples showing γ and k are not separately identifiable.

Part (c): Influenza Outbreak Model

For the British boarding school data with $\delta = 0$, $S(0) = 760$, $I(0) = 3$, $R(0) = 0$, $N = 763$:

OLS initial values:

$$\gamma = 0.0022, \quad r = 0.4469$$

Initial covariance matrix:

$$V = \begin{bmatrix} 3.3117 \times 10^{-10} & 1.3030 \times 10^{-7} \\ 1.3030 \times 10^{-7} & 1.5355 \times 10^{-4} \end{bmatrix}$$

Bayesian parameter estimates:

$$\gamma = 0.0022, \quad r = 0.4487$$

Observational variance comparison: - Bayesian: $\sigma = 19.6142$ - Frequentist: $\sigma = 18.0413$

Chain covariance matrix:

$$V = \begin{bmatrix} 1.9002 \times 10^{-9} & 2.3783 \times 10^{-7} \\ 2.3783 \times 10^{-7} & 2.8680 \times 10^{-4} \end{bmatrix}$$

Figure 6: Chain plots and marginal distributions for influenza model.

Problem 3: Helmholtz Energy Model MCMC

Problem Statement

Consider the Helmholtz energy:

$$\psi(P, \theta) = \alpha_1 P^2 + \alpha_{11} P^4 + \alpha_{111} P^6$$

where P is the polarization on $[0, 0.8]$ and $\theta = [\alpha_1, \alpha_{11}, \alpha_{111}]^\top$.

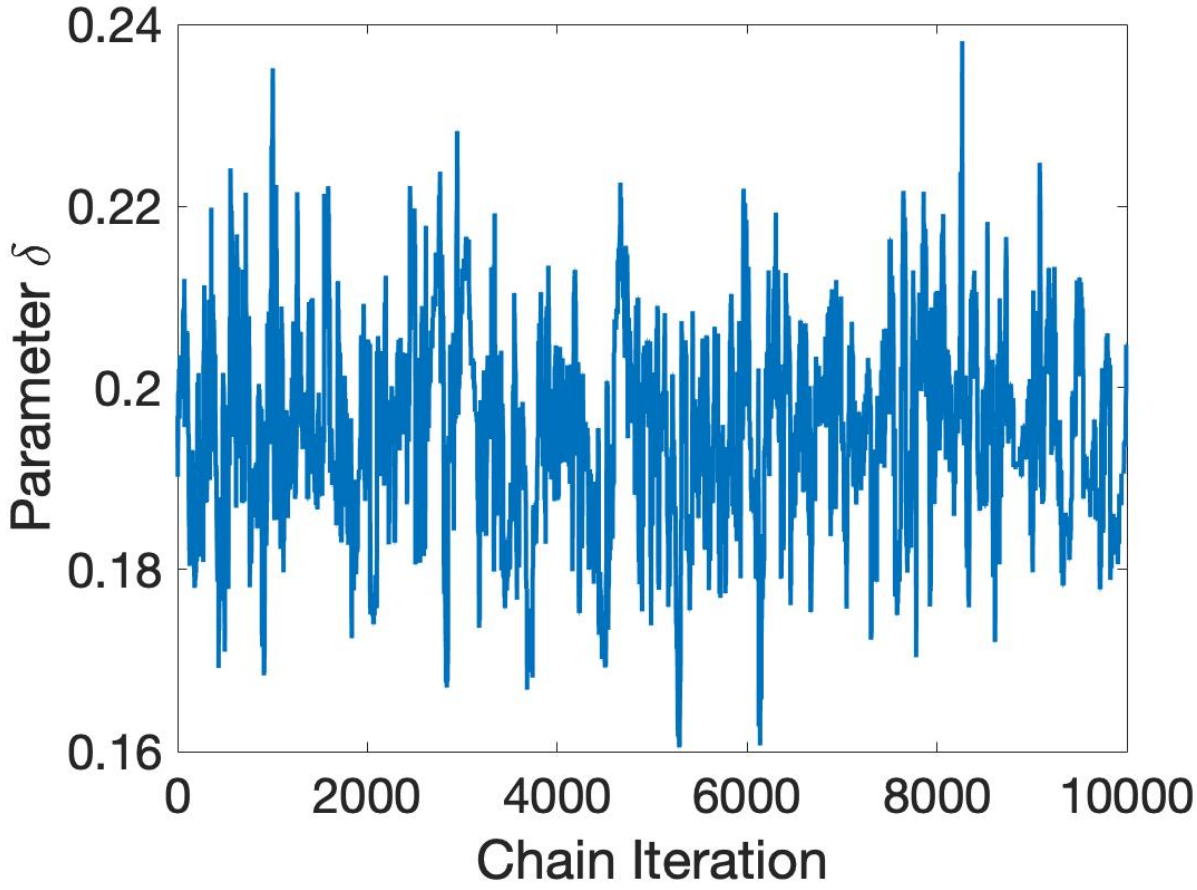

Figure 11: Influenza Chains

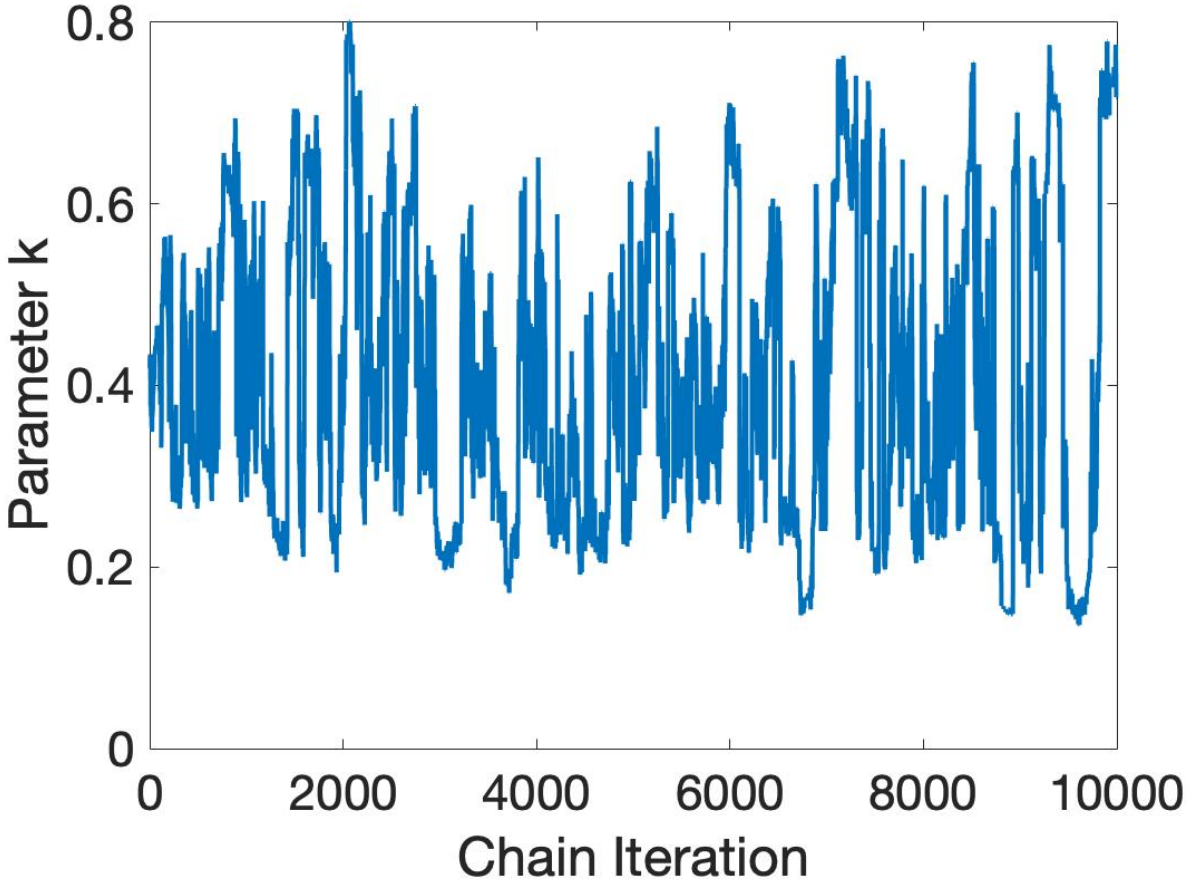

Figure 12: Influenza Distributions

OLS Estimation

Using `fminsearch.m` with the Helmholtz.txt data:

$$\alpha_1 = -382.99, \quad \alpha_{11} = 717.90, \quad \alpha_{111} = 145.27$$

With $n = 81$ data points and $p = 3$ parameters:

$$\sigma^2 = \frac{1}{n-p}\mathbf{R}^\top\mathbf{R} = 12.2068$$

Covariance matrix:

$$V = \begin{bmatrix} 120.01 & -582.35 & 649.33 \\ -582.35 & 3.08 & -3.61 \\ 649.33 & -3.61 & 4.38 \end{bmatrix}$$

Part (a): DRAM Results

Estimated parameters:

$$\alpha_1 = -382.47, \quad \alpha_{11} = 715.6, \quad \alpha_{111} = 147.88$$

Estimated observational variance: $\sigma^2 = 13.4016$

Chain covariance matrix:

$$V = \begin{bmatrix} 111.70 & -532.36 & 588.08 \\ -532.36 & 2.81 & -3.29 \\ 588.08 & -3.29 & 4.01 \end{bmatrix}$$

Part (b): Metropolis Algorithm Results

Estimated parameters:

$$\alpha_1 = -382.87, \quad \alpha_{11} = 716.99, \quad \alpha_{111} = 146.48$$

Estimated observational variance: $\sigma^2 = 13.4187$

Comparison

Both algorithms converge and produce similar results. The marginal posterior distributions from DRAM and Metropolis are nearly identical. All results are close to OLS estimates.

The strong correlation between parameters (evident in pairwise plots) indicates why global sensitivity analysis techniques assuming parameter independence may give misleading results.

Figure 7: Chain plots and pairwise samples using DRAM.

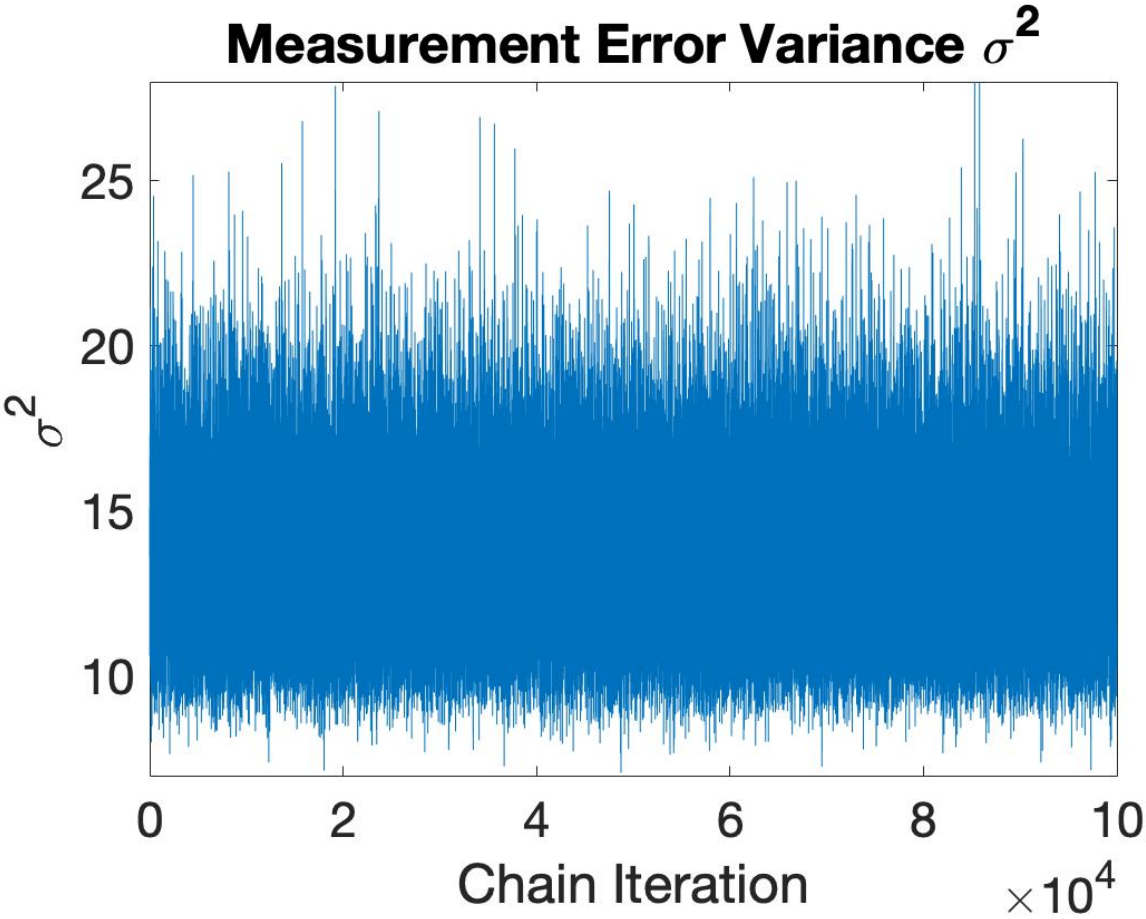

Figure 13: Helmholtz DRAM Chains

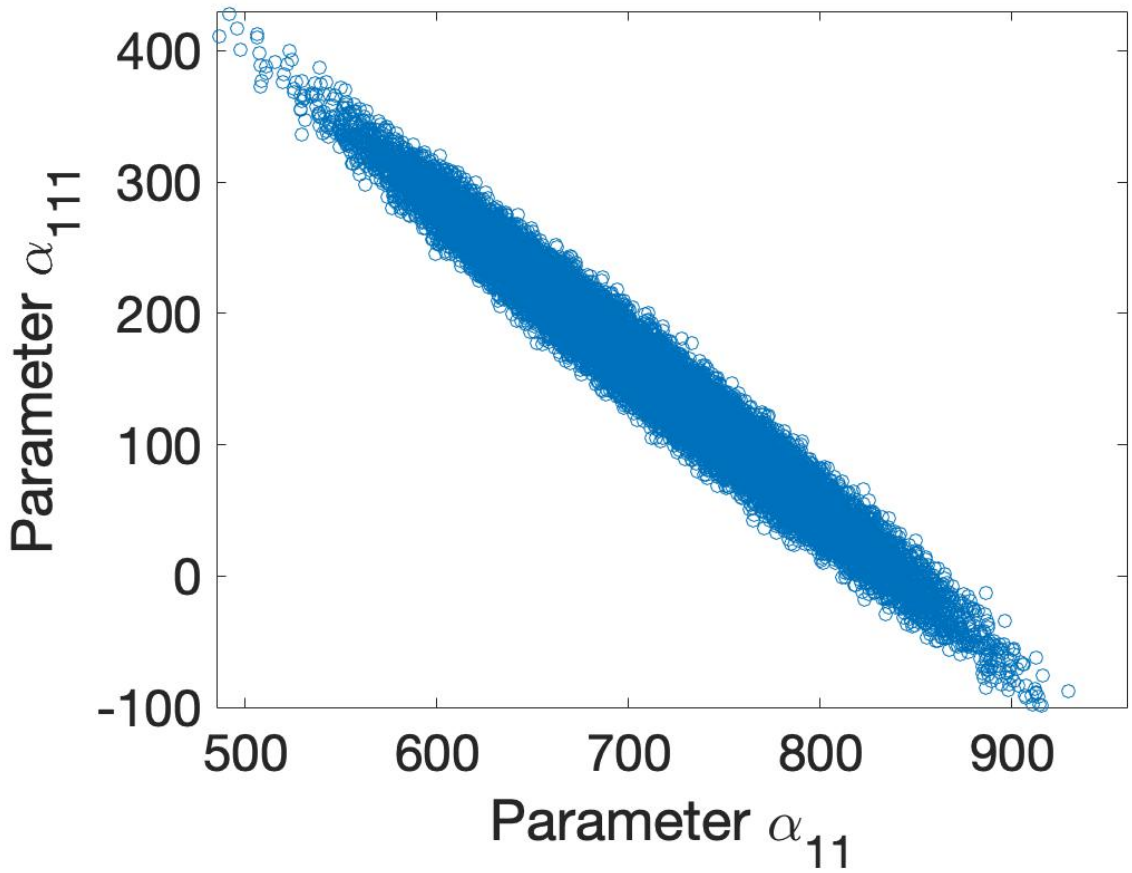

Figure 14: Helmholtz DRAM Pairwise

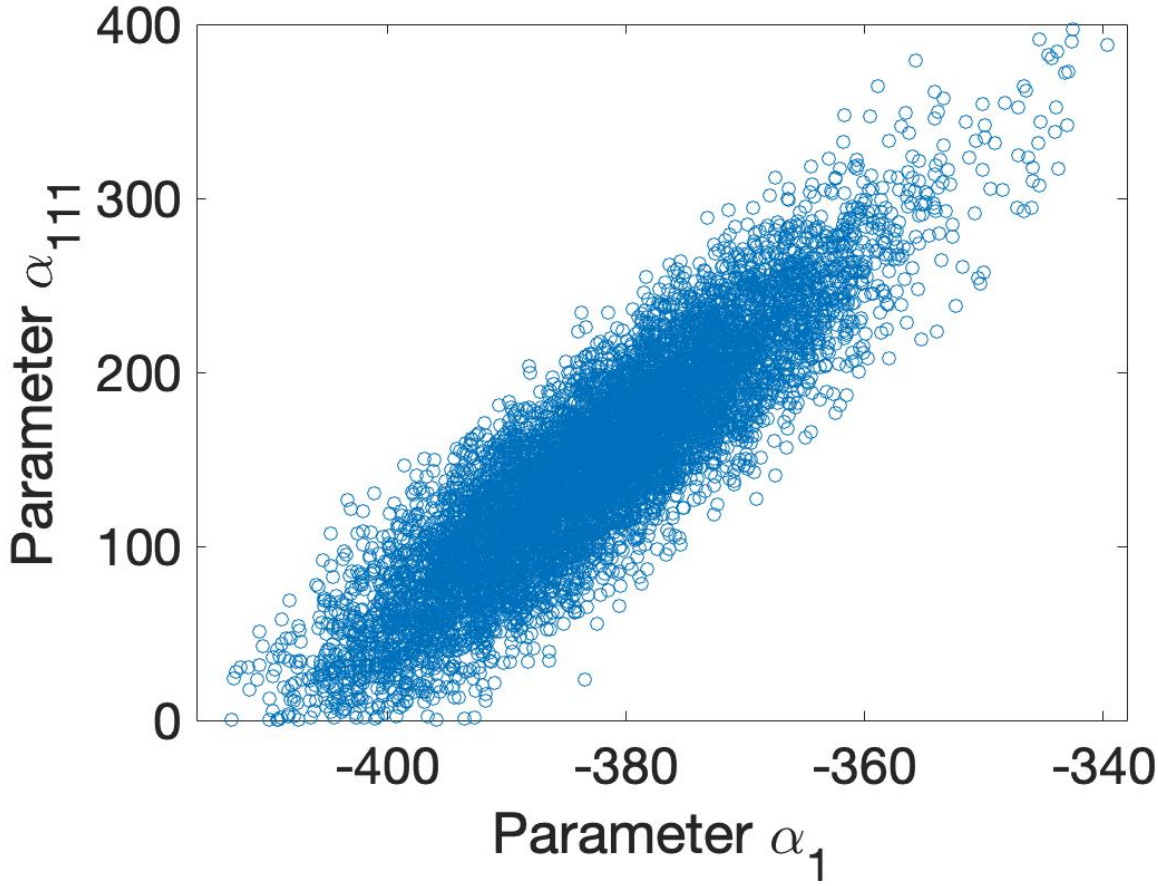

Figure 15: Helmholtz Metropolis Chains

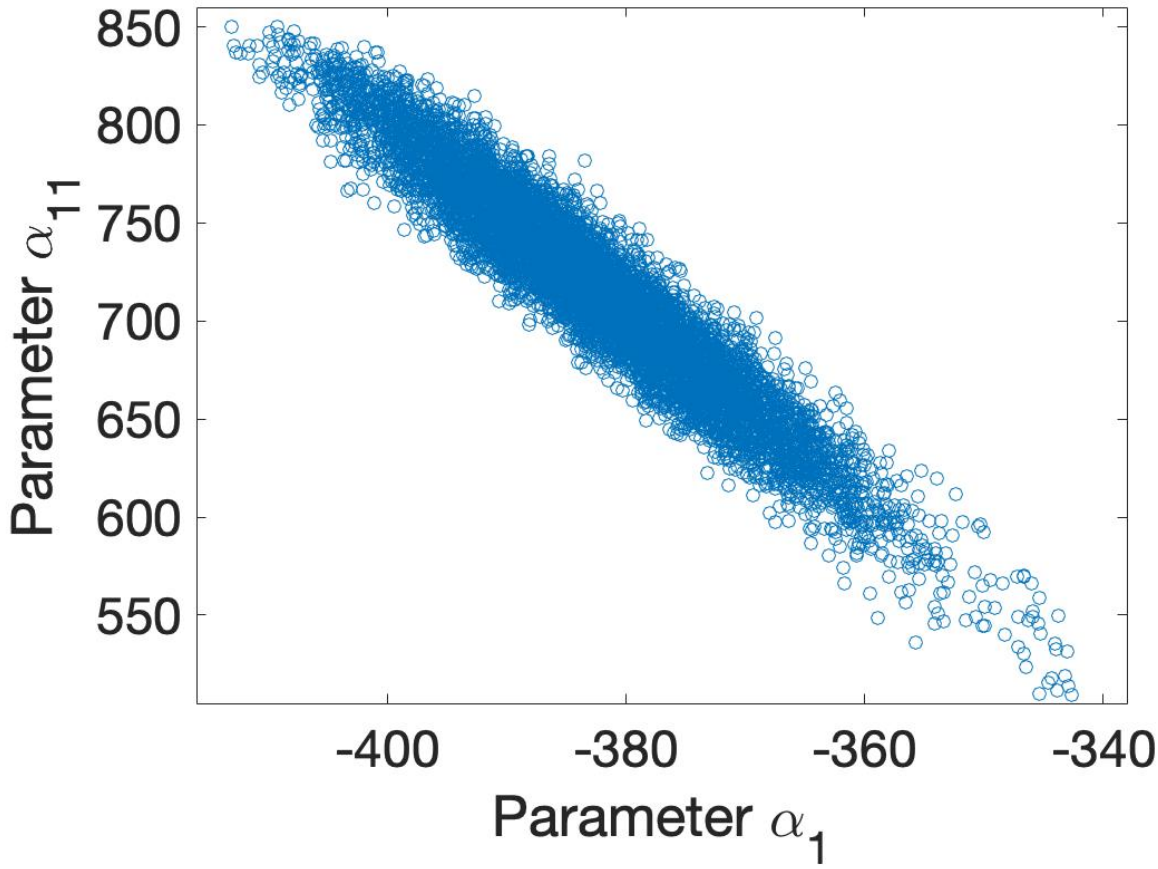

Figure 16: Helmholtz Metropolis Pairwise

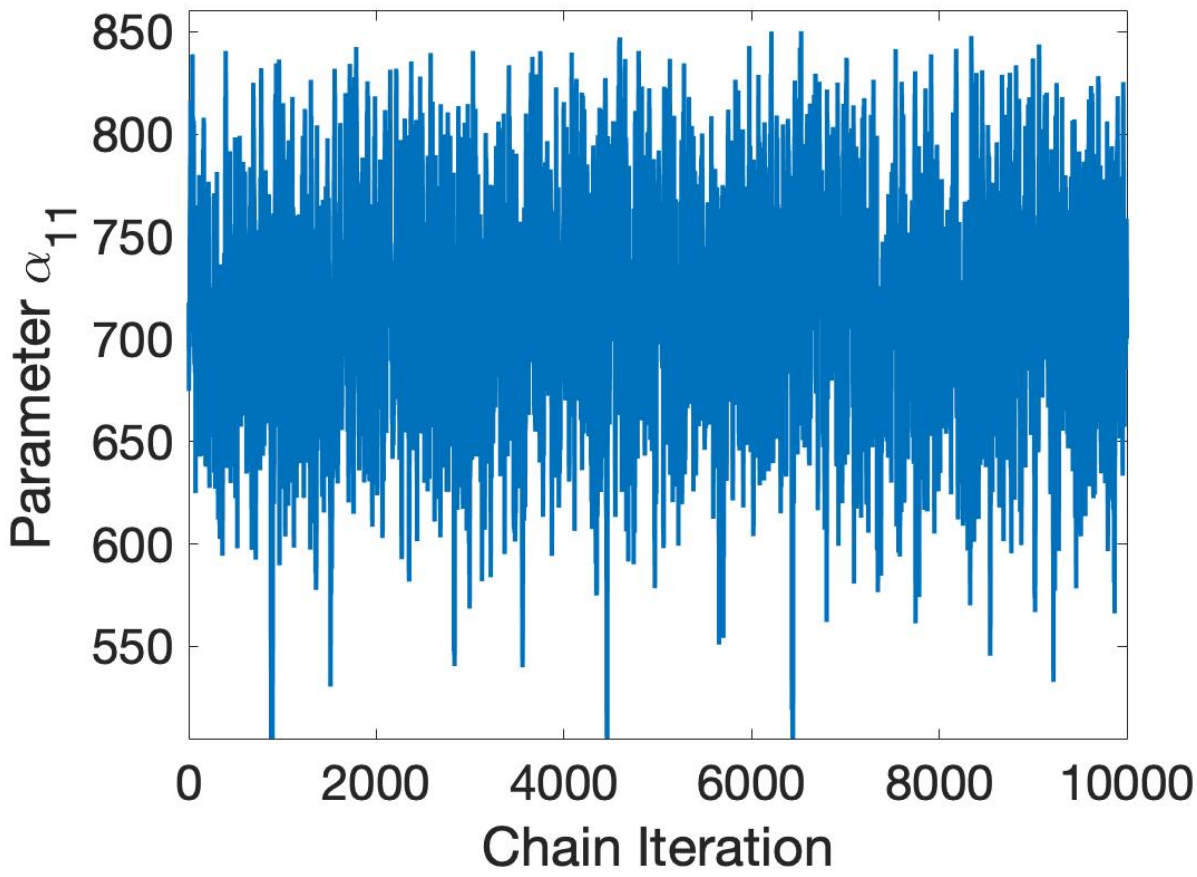

Figure 17: Helmholtz Marginal Distributions

Figure 8: Chain plots and pairwise samples using Metropolis Algorithm.

Figure 9: Marginal posterior distributions comparing DRAM and MCMC.

Code Files

File	Description
Problem1.m / Problem1.py	Heat model MCMC comparison
Problem2.m / Problem2.py	SIR model DRAM analysis
Problem3.m / Problem3.py	Helmholtz MCMC comparison

References

1. Haario, H. et al. (2006). DRAM: Efficient adaptive MCMC. *Statistics and Computing*.
2. Gelman, A. et al. (2013). *Bayesian Data Analysis*, 3rd Edition. CRC Press.
3. Smith, R.C. (2013). *Uncertainty Quantification*. SIAM.